

Notes 5

For each of the following commands include: Definition, Usage/Formula, and Examples.

ls: The ls command is used to list the files and folders in a directory.

Usage / Formula:

```
ls
```

Examples:

```
- ls
- ls Documents
- ls -l**
```

pwd The pwd command shows the current working directory (where you are in the system).

Usage / Formula:

```
pwd
/home/student/Documents
```

cd The cd command is used to change directories (move between folders).

Usage / Formula:

```
cd [directory_name]
```

Examples:

```
- cd Documents
- cd ..
- cd /home/student
```

Include answers to the following questions:

1. **What is a variable?** A variable is a name that stores a value. The value can change.

2. **How do I use a variable?** You assign a value and then use it later. **Example:**

```
name=John  
echo $name
```

3. **What is an environment variable?** An environment variable is a variable that is already set by the system and used by programs. **Example:**

```
echo $HOME
```

4. **What is a user defined variable?** A user defined variable is a variable created by the user. **Example:**

```
age=21  
echo $age
```

5. **What is the root directory?** The root directory is the main directory of the system. Everything starts from here.

Symbol:

```
/
```

6. **What does “Parent Directory” mean?** The parent directory is the folder above your current folder.

Example: If you are in:

```
/home/student/Documents
```

The parent is:

```
/home/student
```

7. **What does “Current working directory” mean?** It is the directory you are currently in.

You can check it with:

```
pwd
```

8. What is an absolute path? Include an example An absolute path is the full path starting from the root directory.

Example:

```
/home/student/Documents/file.txt
```

9. What is a relative path? Include an example A relative path is a path based on your current location.

Example:

```
If you are already in /home/student:  
Documents/file.txt
```

10. What is the difference between “Your home directory” and “The home directory”? Your home directory: Your personal folder **Example:**

```
/home/student
```

The home directory: The general location where all users' home folders are stored.

Example:

```
/home
```